

EX:02	Design and Implement LLM-Based Applications for Question Answering, Text Summarization, and Sentiment Analysis
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Aim:

To design and implement Large Language Model (LLM)-based applications for **Question Answering, Text Summarization, and Sentiment Analysis** using the **Google Gemini API**, and evaluate their performance on real-world textual data.

Procedure:

1. Install the required Python libraries (google-genai).
2. Obtain a Google Gemini API key.
3. Initialize the Gemini client using the API key.
4. Select real-world textual data (news article, product review, or paragraph).
5. Implement a **Question Answering** application by asking questions about the text.
6. Implement a **Text Summarization** application to generate a concise summary.
7. Implement a **Sentiment Analysis** application to classify the text as Positive, Negative, or Neutral with an explanation.
8. Execute the applications using the Gemini model.
9. Compare the generated outputs with the expected results.
10. Record the observations and evaluate the model's performance.

Code:

```

from google import genai

# Initialize Gemini client
client = genai.Client(api_key="AQ.Ab8RN6LIugHudObugXC-oY_vX1B9-
aoMLi2QY7a7aLQjweOI4g")

text = """
Artificial Intelligence is transforming industries by automating repetitive
tasks, improving decision-making, and enabling intelligent applications.
Many companies use AI for healthcare, education, finance, and transportation.
Although AI offers many advantages, ethical concerns such as privacy,
bias, and job displacement remain important challenges.
"""

# -----
# 1. Question Answering
# -----
question = "What are the challenges of Artificial Intelligence?"

qa_prompt = f"""
Read the following text and answer the question.

```

Text:

{text}

Question:

{question}

"""

```
qa_response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents=qa_prompt
)
print("="*60)
print("QUESTION ANSWERING")
print("="*60)
print(qa_response.text)
```

2. Text Summarization

summary_prompt = f"""

Summarize the following text in 3-4 sentences.

{text}

"""

```
summary_response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents=summary_prompt
)
print("\n"+"="*60)
print("TEXT SUMMARIZATION")
print("="*60)
print(summary_response.text)
```

3. Sentiment Analysis

sentiment_prompt = f"""

Analyze the sentiment of the following text.

Classify it as:

- Positive
- Negative
- Neutral

Also provide a brief explanation.

Text:

{text}

"""

```
sentiment_response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents=sentiment_prompt
)
```

```
print("\n"+"="*60)
print("SENTIMENT ANALYSIS")
print("="*60)
print(sentiment_response.text)
```

Output:

```
=====
QUESTION ANSWERING
=====
```

The challenges of Artificial Intelligence mentioned in the text are:

- * privacy
- * bias
- * job displacement

```
=====
TEXT SUMMARIZATION
=====
```

Artificial Intelligence is significantly transforming various industries by automating repetitive tasks, enhancing decision-making, and enabling intelligent applications. Its widespread adoption is evident across sectors such as healthcare, education, finance, and transportation. While AI offers numerous advantages, it also presents critical ethical concerns, including issues of privacy, potential bias, and job displacement, which remain important challenges to address.

```
=====
SENTIMENT ANALYSIS
=====
```

****Sentiment:** Positive**

****Explanation:****

The text primarily highlights the beneficial aspects and capabilities of Artificial Intelligence, such as "transforming industries," "automating repetitive tasks," "improving decision-making," and "enabling intelligent applications." It explicitly states that "AI offers many advantages." While it does acknowledge "ethical concerns such as privacy, bias, and job displacement" as "important challenges," these are presented as issues **within** an otherwise advantageous technology, rather than the technology itself being negative. The overall tone is one of describing AI's powerful and positive impact, with a necessary nod to its complexities.

Result:

The **LLM** was successfully used to implement **Question Answering, Text Summarization, and Sentiment Analysis** applications. The model generated accurate answers, concise summaries, and appropriate sentiment classifications for the given real-world text. The experiment demonstrates that LLMs are effective for a variety of natural language processing tasks and can produce high-quality results with simple prompts.